Beyond the Sound Barrier

An Analysis of Supersonic Air Flow

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Abstract

To revolutionize the aviation industry, supersonics must be studied to improve the efficiency of supersonic aircraft such as the Concorde. Supersonics, however, occupy a distinct physical regime in which compressibility, thermodynamics, and wave phenomena fundamentally alter fluid behavior. This paper develops a simplified treatment of supersonic compressible inviscid flow relevant to supersonic vehicles, beginning with the applied conservation laws derived from the Reynolds Transport Theorem. This treatment is applied to study one and two dimensional wave phenomena such as shocks and regions of heat addition. Finally a pressure measurement technique discussed to highlight how this works findings are used in practice. Together, these topics provide a coherent introduction to the physics governing supersonic flight.

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1. TAKEOFF

It is the morning of October 14th, 1947 and sitting in the bomb bay of a B-29 Bomber lies the Bell XS-1 piloted by Captain Charles Yeager, pictured in Figure 1.1. Despite several broken ribs, World War II veteran and P-51 pilot Captain Chuck finds himself at the forefront of aviation history. At 10:26 A.M. and a speed of 250 mi/h, the brightly painted X-1 drops free from the bomb bay and fires its rocket engine. Producing 6000 lb of thrust, the airplane accelerates and climbs rapidly into the atmosphere. Along the exterior of the plane, shock waves rattle along the wings and at an altitude of 40000 ft the plane levels off. The Mach meter climbs to Mach 1.02 then Mach 1.06 crowning Chuck Yeager as the first pilot to successfully fly faster than the speed of sound. [1] [2]

The X-1's flight sparked dreams of the future applications of supersonic flight. Such dreams culminated in the production of the Concorde, which entered service in 1976 after decades of optimism about bringing supersonics to the commercial market. Developed by Britain and France during the 1950s and 1960s, Concorde demonstrated that commercial supersonic flight was technically feasible, cutting transatlantic travel times drastically.



FIG. 1.1. The world's first manned supersonic airplane, the Bell X-1 Glamorous Glennis, soaring above Muroc Dry Lake.[3]



FIG. 1.2. Concorde, Fox Alpha, Air France on display in the Smithsonian National Air and Space Museum. The first supersonic commercial airliner, which inspires the future of air travel today. [4]

Achieving this performance, however, required operating in a fundamentally different physical regime than before. This regime would prove too harsh for the Concorde, and it was grounded due to the lack of efficiency [4].

Within supersonics, thermodynamic and wave effects not only cannot be ignored, but their properties play vital roles in defining the fluid dynamics present. In the case of the Concorde, not taking such effects into adequate consideration resulted in the inefficiencies that grounded the revolutionary airliner. To bring aviation into the future, we need to study and understand these effects. We will look at introductory examples of the physical phenomena present on a supersonic vehicles, both over the body and through a propulsion system. We will then discuss what measurement techniques are used to study these examples in the real world.

We will begin by familiarizing ourselves with the thermodynamic properties, flow regimes, and the notion of compressibility in Section 2. Building upon those ideas we will introduce compressible flow in Section 3, looking at how such flow can be characterized. We will then apply our findings to wave dynamics in one and two dimensions in Sections 3.2 and 3.3, as well as the effects present within a high Mach number propulsion device in Section 3.4. We will then discuss how our idealized investigations into compressible flow can be useful in real world analysis in Section 5. Finally we will consider the future applications of compressible flow analysis in Section 4.

2. BACKGROUND

The physical regime that we are studying is **supersonic compressible inviscid fluid flow**, but before we dive into understanding its application to supersonic flight we need to define what that regime is. Moreover, we need to define the variables, assumptions, and concepts at play.

2.1. Flow Regimes

Let us tackle this one word at a time, starting with **supersonic**. Supersonic tells us how fast we are moving, that is faster than the speed of sound. We will use what is known as the Mach number to define this mathematically. The Mach number is simply the ratio of the velocity, u, to speed of sound, a, at a point, given by,

$$M = u_n/a_n, \tag{2.1}$$

where n is the point in the flow field, whether that be free stream, ∞, or a local point of interest. [1]

In everyday life we live in the *subsonic* regime, where all velocity vectors in the flow field are such that $M < 1$. The fluid flows smoothly and deflect upstream of a body, such as your car. Subsonic flow is pictured in Figure 2.1a.

This is not our area of concern though; as we accelerate past $M = 1$, we enter the *supersonic* regime. Such flow is pictured in Figure 2.1c and in this regime, the state variables change across the shock wave shown in Figure 2.1d. When supersonic flow occurs, the interactions of the fluid and its physical, thermodynamic, and wave properties change. A shock wave is produced by the presence of a rigid body, across which the streamlines direction changes discontinuously. We will see how this shock wave is produced and how it changes the fluid flow in Section 3.2.3 and beyond.

You may ask yourself, that if the goal is to revolutionize flight times and cross the globe in mere hours, why we don't continue to accelerate, past say $M = 5$. If we were to do so we would land in the *hypersonic* regime. The shock waves move closer to the body's surface, as shown in Figure 2.1e. This results in a huge spike in heat, such that the gas dissociates

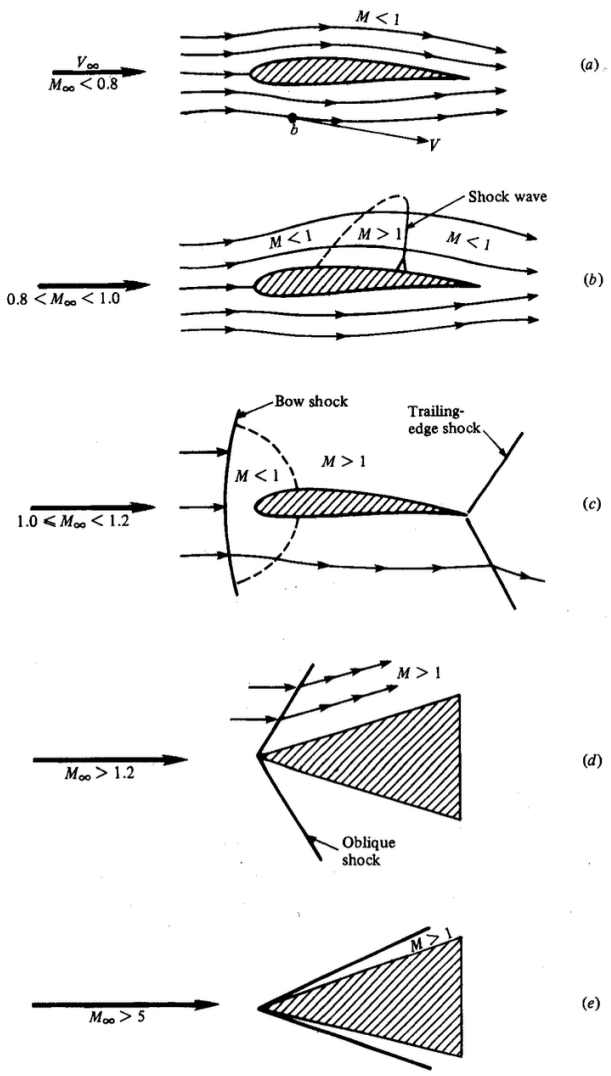


FIG. 2.1. Depiction of different flow over a rigid body (airfoil) Notice that in (a) the flow is subsonic and smooth. While in (d) that the flow is supersonic and discontinuously deflected along the rigid body. Finally in (e) notice how thin the boundary layer is, which produces large amounts of heat and complicates the analysis. [1]

or even ionizes. The aspects of hypersonic flow, its shock waves, chemically reacting fluid, and immense heat add a huge amount of complexity to any analysis. Such analysis is still being done across the physics community today, as an open field of graduate study. For the purpose of the present work we will investigate compressible supersonic flow and mention some hypersonic considerations in Section 4, as a future area of interest.

2.2. Thermodynamics

Before moving onto the next word in our regime of interest, we need to consider the new dependency that arises within supersonics. Due to the high speeds and kinetic energies present, thermodynamics becomes an integral part of the flow, fundamentally linking together fluid dynamics and thermodynamics. To see how, we will look at the assumptions that we need to make and the subsequent simplification of the analysis.

2.2.1. Perfect Gas

The first assumption that we will make is that the fluid is a *perfect gas*, meaning that the average distance between particles is more than 10 molecular distances, so intermolecular forces are very weak and negligible. Therefore, the ideal gas law applies, which one will recall is,

$$\begin{aligned} pV &= \eta RT, \\ pv &= RT, \\ p &= \rho RT, \end{aligned} \tag{2.2}$$

where p is the pressure, v is the specific volume (m^3/kg), and η is the mole-mass ratio $[(kg \cdot mol)/kg]$, R is the specific gas constant, and T is the temperature of the system (K).

We will reference the ideal gas law, in either form, as the equation of state in the subsequent sections. [1] [2] [5]

2.2.2. Heat Capacities

The other assumption that we will make is that our fluid is a calorically perfect gas, meaning that the heat capacities are constant.

The *heat capacity* is the amount of heat, Q, required to raise its temperature, T, given by,

$$C = \frac{Q}{\Delta T} = \frac{(\Delta E - W)}{\Delta T}. \tag{2.3}$$

where W is the work done on the system. If $W = 0$ then there is no volume change, so we can say that the **heat capacity at constant volume**[1], denoted C_V, is

$$C_V = \left(\frac{\Delta E}{\Delta T}\right)_V = \left(\frac{\partial E}{\partial T}\right)_V. \tag{2.4}$$

On the other hand, if the system expands, as they do in everyday life, the object does work on its surroundings hence $W < 0$, so $C > C_V$. To compensate for the lost work, heat is added, and if the pressure surrounding the object is constant, then this value is known as the **heat capacity at constant pressure**[2], denoted C_P and shown in Equation 2.5 .

$$C_P = \left(\frac{\partial E}{\partial T}\right)_P + P\left(\frac{\partial V}{\partial T}\right)_P. \tag{2.5}$$

The second term is the additional heat needed to compensate for the energy lost as work. In our case of gases, this amount of heat is quite significant. [5]

The assumption of a calorically perfect gas (constant heat capacities), allows us to relate different state variables within a supersonic flow. This is primarily done by using the ratio of the different specific heats, the adiabatic exponent, given by

$$\gamma = c_p/c_v. \tag{2.6}$$

To make use of the adiabatic exponent, consider our system as a whole. It has an has an internal energy, E, and an *enthalpy*, $h = E + pv$. [3]

Now, since we assumed that our perfect gas is not chemically reacting, the system is then a thermally perfect gas, where internal energy and enthalpy are functions of temperature only, and where the specific heats at constant volumes and pressure are also functions of temperature. Thus,

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[1] Note that the subscript on the parentheses denotes that volume V is held constant.

[2] Note that the subscript on the parentheses denotes that pressure P is held constant.

[3] One can think of Enthalpy as the amount of energy needed to create the system and to put it into an environment, hence the term pv is the work. Such as creating this paper and moving the air out of the way to place it in front of you.

$$\begin{aligned} E &= E(T), \\ h &= h(T), \\ dE &= c_v dT, \\ dH &= c_p dT. \end{aligned} \tag{2.7}$$

If the specific heats are constant then our system is a *calorically perfect gas*, where

$$\begin{aligned} E &= c_v T, \\ h &= c_p T. \end{aligned} \tag{2.8}$$

By manipulating these equations and making use of Equation 2.2 as well as the definition of enthalpy that,

$$c_p - c_v = R, \tag{2.9}$$

where the specific heats are defined as,

$$\begin{aligned} c_p &= \left(\frac{\partial h}{\partial T}\right)_p, \\ c_v &= \left(\frac{\partial E}{\partial T}\right)_v. \end{aligned} \tag{2.10}$$

.

Dividing Equation 2.9 by c_p and using $\gamma = c_p/c_v$, we can see that,

$$1 - \frac{1}{\gamma} = \frac{R}{c_p}, \tag{2.11}$$

thus,

$$c_p = \frac{\gamma R}{\gamma - 1}, \tag{2.12}$$

and if we do the same process for c_v we find,

$$c_v = \frac{R}{\gamma - 1}. \tag{2.13}$$

Equations 2.12 and 2.13 hold for a thermally or calorically perfect gas; hence, these will be useful in our simplified investigation of supersonic compressible flow. [1]

2.2.3. The Laws and Entropy

Now that we understand the variables within thermodynamics, we can understand what processes change these variables over time under our assumptions.

Suppose for a second that our system is stationary and closed. We let δq be the incremental amount of heat added, while δw is the incremental amount of work done on the system. Due to the molecular motion of the gas, the system has an internal energy E. Then change in energy can be written as:

$$\delta q + \delta w = dE. \tag{2.14}$$

This is the *first law of thermodynamics* and for a given dE there are an infinite number of different processes by which heat and work can be introduced to the system, however the ones of greatest importance to us are as follows:

1. An *adiabatic process*, where no heat is added to or taken away from the system
2. A *reversible process*, where no dissipative actions such as viscous effects, conductivity, or diffusion take place.
3. An *isentropic process* where there is no change in entropy, and is therefore adiabatic and reversible.

In these processes, however, the first law (Eq. 2.14) does not tell us the direction in which the processes will occur. Consider a block of ice sat behind a rocket thruster, for example. Our intuition tells us that when the thruster turns on its exhaust will melt the block of ice. Equation 2.14, however, allows for the ice to get colder and the exhaust gas to get hotter so long as the energy is conserved. This of course does not happen because there is some natural direction in which these processes occur. This is where *entropy* is introduced.

Entropy is the measure of a system's *multiplicity* denoted, Ω, which is explained in Appendix A. For large systems such as our gaseous flow field, the overwhelming dominance of high multiplicity states leads to irreversibility when certain effects are present. This notion that multiplicity, hence entropy, tends to increase is the *second law of thermodynamics*. Entropy is given by the following equation,

$$s \equiv k \ln \Omega, \tag{2.15}$$

where k is the Boltzmann constant which gives entropy, S, units of energy (J/K). We can now say that the second law of thermodynamics is *entropy tends to increase*. Meaning in all processes $ds \geq 0$. An example explaining multiplicity is available in Appendix A. [5]

Using this notion of entropy we can compare two states in our system using,

$$s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1}, \tag{2.16}$$

and

$$s_2 - s_1 = c_v \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1}. \tag{2.17}$$

These two equations allow us to calculate the change in entropy for a calorically perfect gas between two states for either a system of constant pressure or volume. We will use them later to prove the directionality of processes across a shock. Furthermore, for an isentropic process we can use equations 2.16 and 2.17 to find that,

$$\frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1}\right)^{\gamma} = \left(\frac{T_2}{T_1}\right)^{\gamma/(\gamma-1)}. \tag{2.18}$$

Equation 2.18 is important as it relates the state variables in our flow field given an isentropic process which is the case for flows outside the boundary layer in compressible flow. Derivations for equations 2.16, 2.17, and 2.18 are available under the derivations section of Appendix A.

2.3. Distinctions of Fluid Dynamics

We will now move onto the second word in our regime of interest, **compressible**.

The first, and most obvious, distinction we can make is that we are dealing with a gas, air. Thus we can make some assumptions within our fluid mechanics analysis.

We are assuming a compressible flow. Thus the density can change. This complicates the conservation equations throughout the flow, as we will see later. We will define compressibility as,

$$\tau = -\frac{1}{v}\left(\frac{dv}{dp}\right), \tag{2.19}$$

where, v is the volume and p is the pressure. [6]

2.3.1. Inviscid Compressibility

Now we are to tackle the last word, and the last assumption of our regime, **inviscid**.

Viscosity, μ, is the measure of the fluid's resistance to deformation under shear stress. Meaning that it is a measure of the friction forces between the fluid elements. Viscous processes are inherently dissipative, producing entropy and becoming significant in regions where velocity, temperature, or composition change rapidly. Typical examples include boundary layers near solid surfaces, internal flows such as long pipes, and the thin shock layers that form around hypersonic vehicles at high altitude. [6]

As our regime definition may suggest we are assuming an inviscid flow, where viscous stresses, thermal conduction, and diffusion are negligible, setting $\mu = 0$. Although this assumption may appear increasingly restrictive, it is often appropriate over large regions of practical flows where gradients are not extreme. For example, much of the external flow over wings and bodies outside the boundary layer, as well as flows through wind tunnels, rocket nozzles, and turbo machinery blade passages, can be accurately modeled as inviscid. In these cases, key aerodynamic quantities such as pressure distributions, lift, and moments can be predicted with good accuracy despite neglecting dissipative effects. The present work focuses on compressible, inviscid flow, with viscous effects considered only as they appear in the internal structure of shocks, and in hypersonics. [2]

3. COMPRESSIBLE FLOW

Now that we understand what **supersonic compressible inviscid flow** is, we can begin to study it and what happens when we put a plane inside of such flow.

While the ultimate goal would be to determine how to maximize efficiency and thrust capabilities in a supersonic aircraft such as the Concorde, we will leave that to the cutting edge engineers. Our goal will be much more simple. Our goal is to determine the flow

properties ($p, \rho, T, \vec{u}$ etc.) across a region of interest, such as the shaded region in Figure 3.1.

Since the complexity within supersonics grows exponentially as you approach the real world, we will investigate simplified fields of lower dimensions as an introductory exercise. Then discuss how one may study larger systems through the use of measurements.

3.1. Governing Equations

The first order of business is to derive the governing equations within this regime. To do so we will make use of the knowledge that mass, momentum, and energy are conserved throughout the flow field, and what is known as Reynold's transport theorem.

Reynold's transport theorem says that the rate of change of a property for a system equals the change within the control volume plus the net flow of that property across its boundaries; hence the property is conserved. In simpler terms, if a property changes over time in region, than it either changed inside that region, or it moved across the region's boundary. Reynolds Transport Theorem (RTT) is,

$$\frac{d}{dt}\int_{V^*(t)} B(\vec{x},t)dV = \int_{V^*(t)} \frac{\partial B(\vec{x},t)}{\partial t}dV + \int_{S^*(t)} B(\vec{x},t)\vec{u_v}\cdot\hat{n}dS, \tag{3.1}$$

where $B(\vec{x},t)$ is the conserved quantity in the fluid, $V^*(t)$ is the control volume, $S^*(t)$ is the control surface, and $\vec{u_v}$ is the velocity of that surface. $V^*(t)$ may be either fixed in space with the fluid moving through it, or moving with the fluid such that the same fluid particles are always inside. That would be a Lagrangian (material) control volumes respectively.

For the following equations we will treat our control volume as Lagrangian, thus it moves with the fluid so $\vec{u}_v = \vec{u}$ on $S^*(t)$ and we will define our surface element to be $d\vec{S} \equiv \hat{n}dS$.

3.1.1. Continuity Equation

Lets us first apply RTT to mass. Let $B(\vec{x},t) = \rho(\vec{x},t)$, the mass per unit volume. The term on the left of RTT (Eq: 3.1), reduces to zero as there is no change in mass, hence mass conservation. Then with $\vec{u_v} = \vec{u}$, and our defined control surface and volume we have,

$$\frac{\partial}{\partial t}\int_{V^*(t)} \rho dV + \int_{S^*(t)} \rho\vec{u}\cdot d\vec{S^*} = 0, \tag{3.2}$$

where $\vec{u} \cdot \hat{n}$ is the local velocity and ρ is the local density.

This is the mass conservation equation, or the continuity equation within inviscid compressible fluid flow. It says that the change in the control volume over time is equal to the amount of fluid that passes through the control surface in that same time. Colloquially stated as if it is gone (first term), then it left across there (second term).

3.1.2. Momentum Equation

We now apply RTT to the conserved linear momentum in the direction normal to the control volume and surface. First let $B(\vec{x}, t) = \rho\vec{u}$, then Newton's second law gives,

$$\frac{d}{dt}\int_{V^*(t)} \rho\vec{u}dV = \vec{F}_{ext}. \tag{3.3}$$

These external forces can be represented as either body forces, or pressure so we can write,

$$\vec{F}_{ext} = \int_{V^*(t)} \rho\vec{f}dV - \int_{S^*(t)} pd\vec{S}. \tag{3.4}$$

Now to find the left side of Equation 3.4 we apply RTT with $\vec{u}_v = \vec{u}$ and our control volume definition, to find

$$\int_{V^*(t)} \frac{\partial(\rho\vec{u})}{\partial t}dV + \int_{S^*(t)} (\rho\vec{u} \cdot d\vec{S})\vec{u} = \int_{V^*(t)} \rho\vec{f}dV - \int_{S^*(t)} pd\vec{S}. \tag{3.5}$$

This equation represents the conservation of momentum within our flow field and is the inviscid compressible version of the Navier-Stokes equation. With the terms on the left being the change in velocity, the first right hand term being the free body force term, and the final term being the pressure dependent term.

3.1.3. Energy Equation

In order to talk about energy, let us first define the amount of energy per unit mass as $\epsilon = E + u^2/2$, the internal energy summed with the kinetic energy. Then the amount of energy volume can be defined as $B = \rho\epsilon$. Using the first law of thermodynamics we can say that,

$$\frac{\partial}{\partial t}\int_{V^*(t)} \rho\epsilon dV = \int_{V^*(t)} \rho\dot{q}dV + \int_{V^*(t)} \rho(\vec{f}\cdot\vec{u})dV - \int_{S^*(t)} \rho\vec{u}\cdot\hat{n}dS, \tag{3.6}$$

where the first term on the right is the net flow of heat per volume into the control volume, the second term is the net work of body forces on the control volume, and the third term is the net pressure on the control surface. If we now apply RTT to the left side of the equation, and apply our control volume definitions we arrive at the complete energy equation,

$$\int_{V^*(t)} \frac{\partial}{\partial t}\left[\rho\left(e+\frac{u^2}{2}\right)\right]dV + \int_{S^*(t)} \rho\left(e+\frac{u^2}{2}\right)\vec{u}\cdot d\vec{S^*} = \int_{V^*(t)} \rho\dot{q}dV + \int_{V^*(t)} \rho(\vec{f}\cdot\vec{u})dV - \int_{S^*(t)} \rho\vec{u}\cdot d\vec{S^*}. \tag{3.7}$$

Since we are performing a low dimensional and simplified analysis, Equation 3.7 does not include the rate of work done on the fluid inside the control volume by a rotating shaft, $\dot{W}_{shaft}$, the rate of work done by viscous stresses on the control surface, $\dot{W}_{viscous}$, nor does it account for the heat added across the control surface due to thermal conduction or diffusion, $\dot{Q}$. This does not impact our analysis as these effects are not present in our case; in other regimes or cases, however, this equation and its counterparts fail.

3.2. One Dimensional Flow

As previously mentioned, we will apply these multidimensional complex governing equations to simple examples to make some important distinctions. The simplest example is one dimensional flow, where, as the name suggests, all of the fluid elements flow linearly along the x-axis, as pictured in Figure 3.1. Across the shaded region the flow properties change as a function of x as the gas flows, whether due to shock, heat addition or any other process, our goal is to calculate the changes to describe the flow across this region. To do so we will apply the governing equations of mass, linear momentum, and energy to the rectangular control volume, pictured as a dashed line in Figure 3.1.

To make the most simplified analysis we will assume that the left and right hand sides each have an area equal to A perpendicular to the flow. Also, we will assume that the flow is steady, with no body forces present.

Thus the continuity equation (Eq. 3.2) becomes

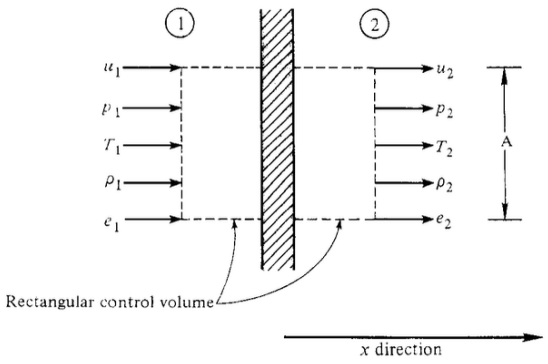


FIG. 3.1. A simplified, one-dimensional rectangular control volume for one-dimensional flow. Flow properties change across the region of interest. This region can involve any process. [1]

$$\int_{S^*(t)} \rho\vec{u} \cdot d\vec{S^*} = 0. \tag{3.8}$$

Since $\vec{u}$ and $d\vec{S^*}$ are parallel but in opposite directions on the left hand side of the shaded region, the integral evaluates to $-\rho_1 u_1 A$. Then on the right side $\vec{u}$ and $d\vec{S^*}$ are parallel and in the same direction hence the integral evaluates to $\rho_2 u_2 A$. Thus,

$$\begin{aligned} -\rho_1 u_A + \rho_2 u_2 A = 0, \\ \rho_1 u_1 = \rho_2 u_2. \end{aligned} \tag{3.9}$$

Moving onto the momentum equation (Eq. 3.5), since we have assumed steady flow with no body forces, the first and third terms are 0, thus we are left with

$$\int_{S^*(t)} (\rho\vec{u} \cdot d\vec{S})\vec{u} = -\int_{S^*(t)} p d\vec{S}. \tag{3.10}$$

Evaluating these surface integrals over the left and right hand side of the control volume, we can see that,

$$\begin{aligned} \rho_1(-u_1 A)u_1 + \rho_2(u_2 A)u_2 = -(-p_1 A + p_2 A), \\ p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2. \end{aligned} \tag{3.11}$$

.

Finally, for the energy equation the first and fourth terms are 0 due to our assumptions and denoting the third term as $\dot{Q}$, the total rate of heat added inside the control volume, we are left with,

$$\int_{S^*(t)} \rho\left(E + \frac{u^2}{2}\right)\vec{u} \cdot d\vec{S^*} = \dot{Q} - \int_{S^*(t)} \rho\vec{u} \cdot d\vec{S^*}. \tag{3.12}$$

Evaluating these surface integrals, once more, we obtain,

$$\dot{Q} - (-p_1 u_1 A + p_2 u_2 A) = -\rho_1\left(E_1 + \frac{u_1^2}{2}\right)u_1 A + \rho_2\left(E_2 + \frac{u_2^2}{2}\right)u_2 A. \tag{3.13}$$

If we rearrange to separate the left and right hand region variables and divide by our one dimensional continuity equation (Eq. 3.9), we get,

$$\frac{\dot{Q}}{\rho_1 u_1 A} + \frac{p_1}{\rho_2} + E_1 + \frac{u_1^2}{2} = \frac{p_2}{\rho_2} + E_2 + \frac{u_2^2}{2}. \tag{3.14}$$

Now since $\dot{Q}$ is the net rate of heat added to the control volume, and $\rho_1 u_1 A$ is the mass flow through the control volume, their ratio is simply q, the heat added per unit mass. And if we apply the definition of enthalpy $h = E + pv$, we can simply further to obtain,

$$h_1 + \frac{u_1^2}{2} + q = h_2 + \frac{u_2^2}{2}. \tag{3.15}$$

Notice how due to our clever choice of geometry and control volumes that our complex equations reduce to intuitive algebraic equations. This will allow us to make easy to understand relations between the state variables.

3.2.1. Speed of Sound Dependence

Now that we have investigated into compressible flow, let us look at what supersonic means and the consequences of the flow being as such. Recall from Equation 2.1, that supersonic depends on the speed of a sound wave. So let us begin by understanding that.

A sound wave is energy being absorbed by surrounding molecules which results in these molecules increasing their mean velocities. As these faster molecules collide with their surroundings, the transfer continues and produces a wave. This wave of energy travels through a medium, such as air, at a velocity that is related to the mean molecular velocity, because

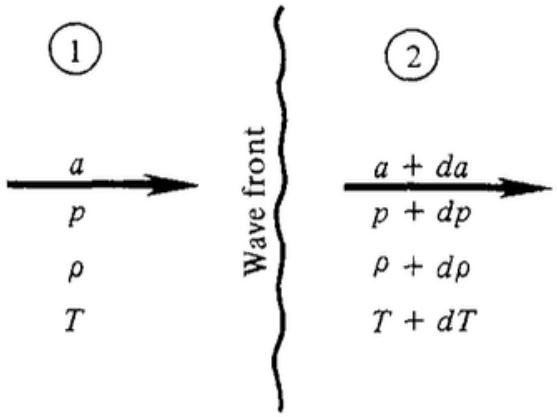


FIG. 3.2. Illustration of a sound wave with infinitesimal changes to flow properties behind the wave front. This wave front is one example of a process that can change the flow variables. [1]

the molecular collisions are propagating the wave. Through the wave the energy increase causes pressure, density, temperature changes which can be measured, by your ear for example in the case of pressure. We will calculate the velocity at which a sound wave propagates through air so that we can apply its effects to our flowfield.

Consider a sound wave moving with a velocity, a, pictured in Figure 3.2. Imagine that we are moving with the wave in space, we see the air in front of us moving toward us at a velocity of a. The flow behind us then would be moving away from us at a velocity of $a+da$, where da is an infinitesimal change in velocity. In fact all of the properties of the wave would change by an infinitesimal amount, as shown in Figure 3.2.

Applying our governing equations between the two states, we find that,

$$\begin{aligned} \rho a &= (\rho + d\rho)(a + da), \\ a &= -\rho \frac{da}{d\rho}, \end{aligned} \tag{3.16}$$

from the continuity equation (Eq. 3.2) and,

$$\begin{aligned} p + \rho a^2 &= (p + dp) + (\rho + d\rho)(a + da)^2, \\ dp &= -2a\rho da - a^2 d\rho, \\ da &= \frac{dp + a^2 d\rho}{-2a\rho}, \end{aligned} \tag{3.17}$$

from the momentum equation (Eq. 3.5). Now we can substitute in the result of our continuity application (Eq. 3.16) to obtain,

$$a = -\rho\left[\frac{dp/d\rho + a^2}{-2a\rho}\right]. \tag{3.18}$$

Solving for a^2, we see that,

$$a^2 = \frac{dp}{d\rho}. \tag{3.19}$$

Now once again consider the physical process that is occurring through the propagation of the sound wave. The changes within the wave are slight, therefore the irreversible effects of friction and thermal conduction are negligible and there is also no heat addition inside the flow. Therefore the propagation of the sound wave must be isentropic. Hence Equation 3.19 is an isentropic relation, thus Equation 2.18 applies.

To show just how fundamental this relation is one can use the definition of density $\rho = 1/v$ and the definition of compressibility (Eq. 2.19), to show that,

$$a = \sqrt{\left(\frac{\partial p}{\partial \rho}\right)_s} = \sqrt{\frac{v}{\tau_s}}. \tag{3.20}$$

This statement says for a fluid to be truly incompressible, the speed through the medium must be infinite. This is to say that no fluid is truly incompressible. Taking our analysis a step further one can show that for an isentropic relation of a calorically perfect gas, the speed of sound is

$$a = \sqrt{\gamma RT}. \tag{3.21}$$

Showing that, for our idealized gas, the speed of sound is a function of temperature only. It is important to keep this in mind throughout our analysis that the speed of sound is a variable, hence the Mach number is dependent on the gas' properties as well, linking fluid dynamics and thermodynamics.

3.2.2. Dimensionless Relations

Now our last tool we need to construct are imaginary ones, the total and stagnation properties of the flow.

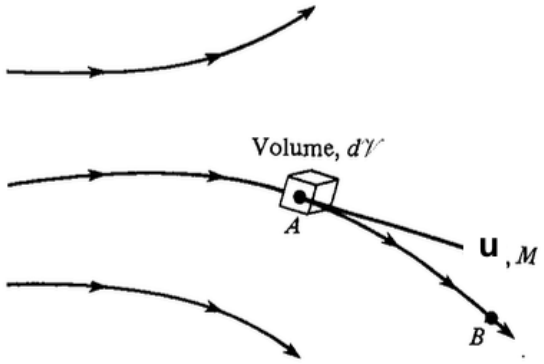


FIG. 3.3. An infinitesimal point in a flow field, with a volume dV moving moving along along a streamline at some velocity u. [1]

First imagine that we take a point within the flowfield such as in Figure 3.3, and we adiabatically bring this fluid element to Mach 1. When this fluid element arrives at Mach 1, it will have some new temperature T^* and hence a new speed of sound, a^*, which can be defined as,

$$a^* = \sqrt{\gamma R T^*}. \tag{3.22}$$

Therefore for any real flow with real values of M and T at some point A, we can define values a^* and T^* to be associated with them.

Similarly, now imagine that we take that same fluid element and we isentropically bring it to rest, stagnating the fluid. The pressure and temperature that the fluid element becomes at $u = 0$, would then be defined as the *stagnation pressure*, p_0, and the *stagnation temperature*, T_0. Once again we can now describe a real moving fluid element at point A, by these imagined values and make relations to them using our fluid dynamic, thermodynamic, and wave state variables.

Some simple distinctions, between the starred quantities and the stagnation properties, are the characteristic Mach number, the stagnation speed of sound and the stagnation density, given below,

$$\begin{aligned} M^* &= V/a^*, \\ a_0 &= \sqrt{\gamma R T_0}, \\ \rho_0 &= p_0/RT_0. \end{aligned} \tag{3.23}$$

Using these relations we can describe the relationships between the real and imagined variables as functions of the Mach number only, for a calorically perfect gas. One of such relations is

$$\frac{a^2}{\gamma - 1} + \frac{u^2}{2} = \frac{\gamma + 1}{2(\gamma - 1)} a^{*2}, \tag{3.24}$$

and is derived in Appendix A. From Equation 3.24, one can derive the ratios of the real and stagnation temperatures and the ratio of the stagnation temperature and the conditions of a^*, p^*, and ρ^* at Mach 1. Our interest however is that if we divide Equation 3.24 by u^2 we find that,

$$\begin{aligned} \frac{(a/u)^2}{\gamma - 1} + \frac{1}{2} &= \frac{\gamma + 1}{2(\gamma - 1)} \left(\frac{a^*}{u} \right)^2 \\ \frac{(1/M)^2}{\gamma - 1} &= \frac{\gamma + 1}{2(\gamma - 1)} \left(\frac{1}{M^*} \right)^2 - \frac{1}{2} \end{aligned} \tag{3.25}$$

$$M^2 = \frac{2}{[(\gamma + 1)/M^{*2}] - (\gamma - 1)}.$$

Equation 3.25 gives us a direct relation between M and M^*. This relation is important as qualitatively M^* acts in the same fashion as as M, except when M goes to infinity as M^* approaches $\sqrt{(\gamma + 1)/(\gamma - 1)}$. Furthermore using this equation we can solve for the state variables across our region of interest as functions of the Mach number alone.

The reason we need these variables is that now, at any point within a compressible flow, we may define a corresponding set of total (or stagnation) properties by imagining that the fluid is brought to rest through an isentropic process.

The actual flow itself does not need to be isentropic between points. Even in the presence of shocks, viscous effects, or heat transfer, stagnation properties can still be defined locally using isentropic relations.

As a result, each point in a general flowfield is associated with its own stagnation state. If the flow between two points is irreversible or non-adiabatic, these stagnation properties will differ from one location to another. However, in the special case of an isentropic flow, the stagnation properties remain constant throughout the field. This invariance makes total conditions particularly valuable, as they provide conserved quantities that greatly simplify the analysis of compressible flows.

3.2.3. Normal Shocks

Now within our one dimensional simplified analysis, lets consider the wave dynamics present within the flow, specifically shocks, stated as an example in Section 3.2. Imagine that the airfoils in Figure 2.1 are now flat plates flying through the sky, and recall that the flow consists of individual molecules. When the molecules impact the face of the plate they change molecular energy and momentum. Just as in our discussion of sound waves the random motion of the molecules communicates this change in energy to other regions of the flow through propagation at the speed of sound, a. This includes the presence of the body directly upstream. When the speed of the flow is supersonic such that $u_\infty > a_\infty$, the sound waves can no longer propagate upstream, since the plate is moving faster than they can communicate its presence, thus they combine over a short distance ahead of the body as a shock wave.

First consider a shock in one dimension or a *normal shock*. This wave is perpendicular to the flow as sketched in Figure 3.4. We will assume that we know all wave properties ahead of the shock and want to determine all properties after the shock wave. Additionally, assume there is no heat addition across the shock making the process adiabatic. Therefore equations 3.9 through 3.15 in conjunction with our thermodynamic relations of a calorically perfect gas (Eq. 2.2 and $H = c_p T$), give us five unknowns that we can solve for algebraically.

The first relation that we will make much use of is the *Prandtl relation* (which one can derive from Equation 3.9, 3.11, 3.15, and 3.21), given by Equation 3.26,

$$a^{*2} = u_1 u_2. \tag{3.26}$$

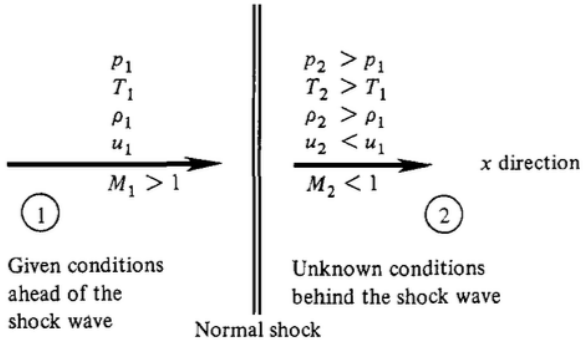


FIG. 3.4. A diagram of a normal shock wave in one dimensional flow, across which all of the variables change. [1]

From the Prandtl relation we can see that,

$$1 = \frac{u_1}{a^*}\frac{u_2}{a^*} = M_1^* M_2^*, \qquad M_2^* = \frac{1}{M_1^*}. \tag{3.27}$$

Meaning behind the normal shock is always subsonic. This is a general statement, not just limited to a calorically perfect gas. If we however apply Equation 3.25 to the Prandtl relation, which does require that we deal with a calorically perfect gas we obtain,

$$M_2^2 = \frac{1 + [(\gamma - 1)/2]M_1^2}{\gamma M_1^2 - (\gamma - 1)/2}. \tag{3.28}$$

Equation 3.28 shows that for a calorically perfect gas with a constant value of γ, the Mach number behind the shock is a function of the upstream Mach number. We can then solve each of the downstream wave properties as functions of the upstream Mach number only. By combining Equation 3.9 with the Prandtl relation (Eq. 3.26), we obtain

$$\frac{\rho_2}{\rho_1} = \frac{u_1}{u_2} = \frac{u_1^2}{u_1 u_2} = \frac{u_1^2}{a^{*2}} = M_1^{*2}. \tag{3.29}$$

By solving Equation 3.25 for M^* and substituting that in we obtain,

$$\frac{\rho_2}{\rho_1} = \frac{u_1}{u_2} = \frac{(\gamma + 1)M_1^2}{2 + (\gamma - 1)M_1^2}. \tag{3.30}$$

This gives the ratios of density and velocities such that we can solve for the downstream properties as functions of upstream Mach number only.

To solve for the pressure ratio, we return to our momentum equation (Eq. 3.11), and combine it with the continuity equation (Eq. 3.9) to see that,

$$p_2 - p_1 = \rho_1 u_1 (u_1 - u_2) = \rho_1 u_1^2 \left(1 - \frac{u_2}{u_1}\right). \tag{3.31}$$

Dividing by p_1 and substituting in Equation 3.29 for u_2/u_1, we obtain,

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1}(M_1^2 - 1). \tag{3.32}$$

Finally from our pressure ratio (Eq. 3.32) we can obtain the temperature ratio by referring to our ideal gas equation, $p = \rho RT$, hence,

$$\frac{T_2}{T_1} = \frac{p_2}{p_1} = \frac{\rho_1}{\rho_2}. \tag{3.33}$$

Thus substituting the pressure ratio and the density ratio into this relation we obtain,

$$\frac{T_2}{T_1} = \frac{h_2}{h_1} = \left[1 + \frac{2\gamma}{\gamma + 1}(M_1^2 - 1)\right]\left[\frac{2 + (\gamma - 1)M_1^2}{(\gamma + 1)M_1^2}\right]. \tag{3.34}$$

We now have relations to solve for the flow fields properties before and after the normal shock wave. Notice that for a calorically perfect gas with a given γ, each property is a function of M_1 only. This is a major distinction in fluid dynamics. For inviscid compressible supersonic flow, the flow properties depend only on the Mach number, which is a thermodynamic property of the fluid present. This link between thermodynamics and fluid mechanics is in stark contrast of incompressible subsonic flow, where thermodynamic effects are assumed to be negligible.

Notice however that Equation 3.27, which stated that the flow behind the shock wave is always subsonic also holds for $M_1 < 1$. Therefore to show that these ratios have physical meaning only when $M > 1$, we must use the second law of thermodynamics. From Equation 2.16, with our pressure and temperature ratios we have,

$$s_2 - s_1 = c_p \ln\left[\left[1 + \frac{2\gamma}{\gamma + 1}(M_1^2 - 1)\right]\left[\frac{2 + (\gamma - 1)M_1^2}{(\gamma + 1)M_1^2}\right]\right] - R \ln\left[1 + \frac{2\gamma}{\gamma + 1}(M_1^2 - 1)\right]. \tag{3.35}$$

This equation demonstrates that the change in entropy across the shock wave is also a function of M_1 only. More importantly it shows that,

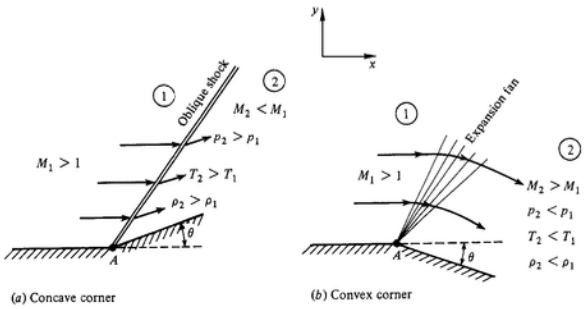


FIG. 3.5. Supersonic flow over a corner. As the flow is forced into itself as shown in (a), a shock wave is formed. The flow properties change, and the flow slows down. The rigid body is angled into the flow, by an angle of θ. [1]

$$\begin{aligned} M_1 < 1, s_2 - s_1 < 0, \\ M_1 = 1, s_2 - s_1 = 0, \\ M_1 > 1, s_2 - s_1 > 0. \end{aligned} \tag{3.36}$$

We know from the second law, that entropy must increase. Thus, the case in which the upstream Mach number is subsonic is physically impossible, which in turn tells us that across a normal shock wave, the pressure, density, and temperature increase, whereas the velocity decreases and the Mach number decreases to a subsonic value.

3.3. Oblique Relations

Now lets expand our analysis to two dimensions, so imagine now that the airfoils in Figure 2.1, are flat triangles with the point facing upstream.

When supersonic flow is turned onto itself as shown in Figure 3.5a, an oblique shock wave is produced. At point A, this surface is deflected upward by an angle θ; therefore the flow streamlines are deflected upward toward the bulk of the flow above the surface. This change in flow direction takes place across a shock wave which is oblique to the free-stream conditions. All of the flow streamlines experience this same oblique deflection angle and as such the downstream flow streamlines are uniform and parallel.

Before we can study oblique shock, however we must first understand where these waves come from. It will come as no surprise that oblique waves are created by the same physical

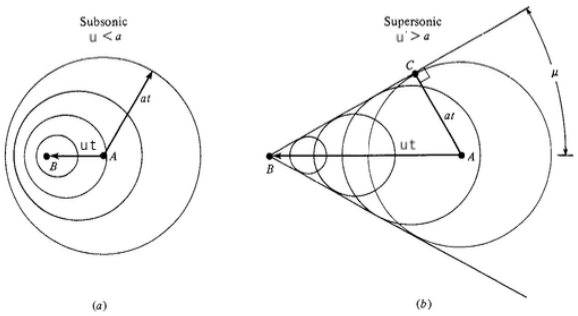


FIG. 3.6. The propagation of disturbances in (a) subsonic and (b) supersonic flow. As the beeper travels it emits sound waves. When the flow is supersonic, a mach wave forms that is tangent to the emitted sound waves. [1]

mechanism as we saw in Section 3.2.3 and in Section 3.2.1 for sound waves, that is, disturbances which propagate by molecular collisions at the speed of sound. We now however are considering such propagation in two dimensions.

To visualize this, consider a moving point source of sound disturbances in a gas, pictured in Figure 3.6. The "beeper" is continually emitting sound waves as it moves through the stationary gas. If the beeper is moving slower than the speed of sound, as shown in Figure 3.6a, then as the beeper travels it will always be inside the family of circular sound waves which are propagating at a speed of a out radially. However, if beeper is traveling at some speed faster than the speed of sound. When it emits a sound wave at point A, from that point the sound wave propagates out at a rate of at. In that same amount of time the beeper has moved an amount ut to point B. Furthermore, on its way to point B, it has emitted more sound waves, sound as smaller circles in Figure 3.6b. The beeper is now always outside of these circles and moving ahead of the wave fronts. Moreover, these wave fronts form a disturbance given by the line BC, which is tangent to the family of circles. This line is known as the *Mach wave* and the angle that it makes (given by ABC) is the *Mach angle*, μ, defined as,

$$\mu = \sin^{-1} \frac{1}{M}. \tag{3.37}$$

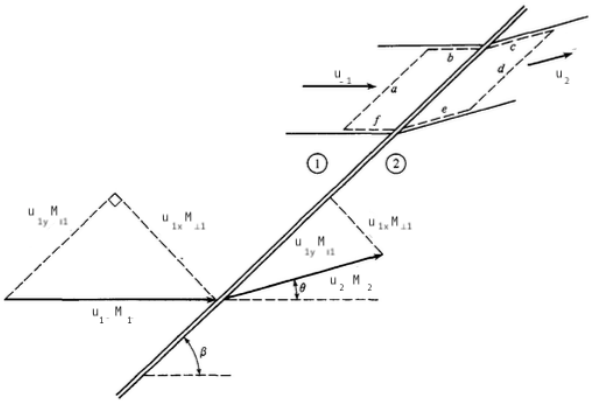


FIG. 3.7. Oblique shock wave geometry. Our triangular body creates a shock at an angle β, which deflects the flow behind the shock by an angle of θ. We decompose the velocity vectors such that our control volumes are parallel to them. As shown by parts a through f. [1]

3.3.1. Oblique Shock Waves

When the disturbance in the flow is more than just a sound emitting point source, such as our interest of the front of a plane flying through the upper atmosphere, the wave front becomes stronger than a Mach wave. The strong disturbance coalesce into a oblique shock at an angle of β to the free streamlines, where $\beta > \mu$. A Mach wave can be thought of as a infinitely weak shock wave. A weak shock wave is one in which causes the flow behind the shock the least changes, and has the smallest deflection angle, θ. Consider our new, stronger oblique shock wave pictured in Figure 3.7.

The upstream velocity is u_1 and is horizontal, the corresponding Mach number is M_1. The oblique shock wave produced is at an angle to the horizontal of β. Behind the shock, the flow is deflected toward the shock by the flow-deflection angle θ, and has a velocity and Mach number u_2 and M_2, respectively. We will apply our governing equations from Section 3.1 the same way that we applied them to one dimensional flow, now considering the normal and tangential components of the velocities.

Same as before we can make a clever choice of our control volume, as shown by the dashed lines a through f, to reduce our integrals to simple algebraic equations.

We first apply Equation 3.2 to the control volume. The surface integral evaluated over

is evaluated over faces a and d only, as the rest of the faces are parallel to the velocity, and hence contribute nothing to the surface integral ($\vec{u} \cdot d\vec{S} = 0$). Thus our continuity equation for an oblique shock wave is,

$$\rho_1 u_1 = \rho_2 u_2. \tag{3.38}$$

Continuing onto our momentum considerations using Equation 3.5, we need to consider two components. First the tangential component, with no body forces applied to the control volume yields,

$$(-\rho_2 u_{1x}) u_{1y} + (\rho_2 u_{2x}) u_{2y} = 0. \tag{3.39}$$

If we divide by our continuity result, we find that $u_{1y} = u_{2y}$, meaning that the tangential component of the flow velocity is conserved across the oblique shock wave. Moving our attention to the normal component of the momentum equation, we find that,

$$p_1 + \rho_1 u_{1x}^2 = p_2 + \rho_2 u_{2x}^2. \tag{3.40}$$

Finally applying our energy equation (Eq. 3.7) to this control volume, we find that,

$$-(-p_1 u_{1x} + p_2 u_{2x}) = -\rho_1 \left(e_1 + \frac{u_1^2}{2} \right) u_{1x} + \rho_2 \left(e_2 + \frac{u_2^2}{2} \right) u_{2x}. \tag{3.41}$$

Subbing in for enthalpy,

$$h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2}, \tag{3.42}$$

and then,

$$h_1 + \frac{u_{1x}^2}{2} = h_2 + \frac{u_{2x}^2}{2}, \tag{3.43}$$

for a steady adiabatic flow with no body forces, where $u^2 = u_x^2 + u_y^2$ and $u_{1y} = u_{2y}$.

Notice that Equations 3.38, 3.40, and 3.43 are identical in form to the equations derived for normal shock relations. Furthermore, the same algebra applied to these equations will lead to identical expressions for changes across an oblique shock in terms of the *normal* component of the upstream Mach number, $M_{\perp 1}$. Meaning that for an oblique shock wave with,

$$M_{\perp 1} = M_1 \sin \beta, \tag{3.44}$$

we have for a calorically perfect gas,

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma+1)M_{\perp 1}^2}{2+(\gamma-1)M_{\perp 1}^2}, \tag{3.45}$$

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma+1}(M_{\perp 1}^2 - 1), \tag{3.46}$$

$$M_{\perp 2}^2 = \frac{M_{\perp 1}^2 + [2/(\gamma-1)]}{[2\gamma/(\gamma-1)]M_{\perp 1}^2 - 1}, \tag{3.47}$$

and

$$\frac{T_2}{T_1} = \frac{p_2}{p_1}\frac{\rho_1}{\rho_2}. \tag{3.48}$$

We can see that for a normal shock wave these relations are functions of only M_1 but for an oblique shock wave they are a function of both $M_\perp 1$ and β. The normal shocks are a special case where $\beta = \pi/2$. Note that behind the shock M_2 can be found from $M_{\perp 2}$ and the geometry of Figure 3.7 as,

$$M_2 = \frac{M_{\perp 2}}{\sin\beta - \theta}. \tag{3.49}$$

The key takeaway is that if the Mach numbers are known, then we can solve for any and all state variables of interest, when it comes to shock waves.

3.4. Ramjet Operation and Heat

Up until this point we have considered the flow over the exterior of a supersonic object, let us now consider propulsion technology and the case of the **ramjet engine**.

At flight speeds above approximately Mach 3, propulsion cycles no longer require rotating components such as compressors to raise the pressure. Instead, compression can be achieved through geometric area changes in the inlet and diffuser upstream of the combustor. Eliminating core rotating machinery allows the engine to operate at higher maximum cycle temperatures, since the temperature limits imposed by turbine materials are no longer a constraint. Under these conditions, it becomes more efficient to preserve supersonic flow

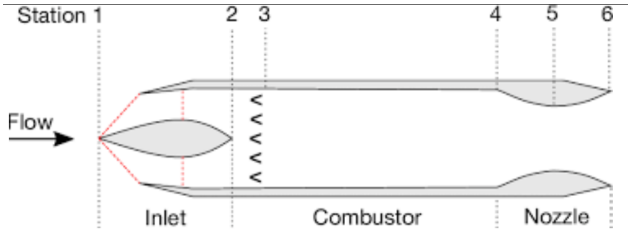


FIG. 3.8. A schematic internal structure of a ramjet, which we will model as one dimensional, with heat addition in the combustor. [7]

throughout the engine and to introduce heat via combustion at supersonic velocities, so a ramjet is used.

The ramjet is an air-breathing engine that operates without moving parts and achieves air compression through internal geometry only. It's purpose to be used at flight speeds near or above Mach 3, where using gas-turbine engines with rotating parts becomes increasingly inefficient. One should note however that the ramjet does not operate from takeoff, rather it is an alternate engine that takes over propulsion responsibilities once a specific velocity is reached.

Now reconsider Figure 3.1, in previous sections we assumed that the changes in flow properties across the shaded region were due to a shock wave. Another source of change is heat addition such as the fuel-air combustion process within a ramjet, pictured in Figure 3.8.

Reconsider our governing equations for one dimensional flow. For a calorically perfect gas we can solve Equation 3.15 for q to find,

$$\begin{aligned} q &= \left(c_p T_2 + \frac{u_2^2}{2}\right) - \left(c_p T_1 + \frac{u_1^2}{2}\right) \\ q &= c_p T_{02} - c_p T_{01} = c_p (T_{02} - T_{01}), \end{aligned} \tag{3.50}$$

where we used the stagnation temperature. Equation 3.50 clearly states that the effect of heat addition is to directly change the stagnation temperature of the flow. Let us find the relations between properties before and after the heat addition as functions of the Mach numbers same as before.

From our momentum equation (Eq. 3.11) and the relation of velocity to Mach number we obtain,

$$\frac{p_2}{p_1} = \frac{1+\gamma M_1^2}{1+\gamma M_2^2}. \tag{3.51}$$

Using this ratio and the relation of a perfect gas, we obtain,

$$\frac{T_2}{T_1} = \left(\frac{1+\gamma M_1^2}{1+\gamma M_2^2}\right)^2 \left(\frac{M_2}{M_1}\right)^2, \tag{3.52}$$

and

$$\frac{\rho_2}{\rho_1} = \left(\frac{1+\gamma M_2^2}{1+\gamma M_1^2}\right)^2 \left(\frac{M_1}{M_2}\right)^2. \tag{3.53}$$

Finally, we can determine the change in entropy using Equation 2.16, which in this case is a direct result of the heat addition and thermal properties as,

$$s_2 - s_1 = c_p \ln\left[\left(\frac{1+\gamma M_1^2}{1+\gamma M_2^2}\right)^2 \left(\frac{M_2}{M_1}\right)^2\right] - R\ln\left(\frac{1+\gamma M_1^2}{1+\gamma M_2^2}\right). \tag{3.54}$$

We can see that no matter the effect that causes the change in properties across our shaded region in Figure 3.1, the change can be represented as a function of the Mach numbers only. [7]

This is a interesting result, as we can solve for a solution for heat addition for any given amount of heat, q. For a given q, T_{02} can be obtained from Equation 3.50 and using that value we can solve for the downstream Mach number M_2 from which we can make use of the above equations for the properties. So for example if we knew the amount of heat injected into our combustor within our ramjet, and how fast we are moving, we could easily solve for the conditions at the nozzle and behind the supersonic vehicle.

4. HYPERSONIC FLOW

We are now equipped with the tools to study inviscid compressible flow, such as the flow over and through a supersonic flying vehicle. Let us now however move our attention to the future. If the goal of the future of aviation is to bring about more efficient travel around the world and beyond our atmosphere, we must travel faster. As such we must concern ourselves with the higher velocity regime, hypersonics.

Hypersonics is a regime in which the upstream Mach number exceeds $M_1 \geq 5$. This however is not 'magic cutoff' where the flow properties instantaneously change from $M_1 =$

4.99 to $M_1 = 5.01$. Rather it is conventional for a Mach number at which many of the phenomena that define hypersonic flow appear.

The only distinction that can be made within this work is that if we take our governing equations and solved ratios of state variables to the limit of $M_1 \to \infty$, for a calorically perfect gas we can obtain some upper bound hypersonic shock relations. This is left as an exercise to the reader.

Due to the phenomena present within hypersonics our assumptions to break down. Firstly, as the Mach number increases beyond Mach 5, the flow becomes nonlinear, and viscous effects start to become too large to assume as negligible compared to other flow parameters. Furthermore, at hypersonic speeds the flow deflection angle approaches the angle of body within the flow, creating thin shock layers. These layers along with the kinetic energy of the gas produce very large temperature and pressure gradients, which in turn change the gas dynamics within the flow field. For our analysis we assumed that our flowfield was comprised of a calorically perfect non-reacting gas. In the hypersonic regime, however, the temperature gradients are sufficient enough to ionize the gas and cause chemical reactions. This changes many properties within the flow, breaking most of the assumptions made in this work. Future analysis would require we apply statistical mechanics to analyze such a large system of reacting particles moving at high speed.

If we wanted to study the operation of a **scramjet** for example, which is a propulsion method that is designed to operate beyond Mach 5, we would have to consider much more factors within the flow. This field of study is known as *high temperature gas dynamics* within the study of hypersonics. Reference [7] is a great resource for beginning such research.

Hypersonics flow analysis demands a coupling of fluid dynamics, thermodynamics, and wave phenomena at a level that extends well beyond the scope of this work. Nonetheless, the concepts developed here provide the essential groundwork upon which modern hypersonic theory is built, and they form the intellectual bridge between classical compressible flow and the frontier challenges that continue to define hypersonic research today.

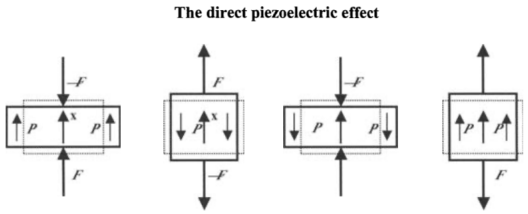


FIG. 5.1. A schematic of the piezoelectric effect in a spliced quartz plate. A force, F, changes the electric polarization, P, allowing us to quantify that force by measuring the change in the polarization and charges.[8]

5. MEASUREMENT TECHNIQUES

One may be wondering, that due to our assumptions and constraints outlined throughout this paper, it is hard to apply these findings directly to the study of supersonics despite very specific conditions. These findings can, however, be used as upper and lower bound limits in real world flows that can be measured through various measurement techniques. We will discuss how real world physicists measure pressure in flow fields through the use of the piezoelectric sensor.

5.1. Pressure Measurements

Piezoelectric sensorics are used to study the pressure changes present through a flow field. They make use of what is known as the piezoelectricity, and the piezoelectric effect.

Piezoelectricity is a linear electromechanical interaction between the mechanical and the electrical state in crystals. Within piezoelectricity, the piezoelectric effect is present when a mechanical deformation of the piezoelectric material produces a proportional change in the polarization. That is, when an electric charge appears on opposite faces of piezoelectric materials when they are acted upon by a mechanical force. A quartz plate cut in half shows the piezoelectric effect in Figure. 5.1. The acting force, F, deforms the quartz plate and produces electric polarization, P. [8]

We can make use of the piezoelectric effect to measure the pressure force in a supersonic aircraft, by measuring the mechanical stress imposed on the material versus the electric changes. To do so we must rely on the Maxwell equations, specifically,

$$\nabla \cdot \vec{D} = \rho_f, \tag{5.1}$$

which tells us that the divergence of the electric displacement field, $\vec{D}$, is equal to the free charge density, ρ_f [9]. In other words, free charges create electric displacement field lines. Those field lines can be described as,

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}. \tag{5.2}$$

Here, $\vec{P}$ is the polarization of the density of the material, and is proportional to the electric field. Which we said earlier is induced by mechanical stress via the piezoelectric effect. Thus we can say,

$$D_i = d_{ijk}\sigma_{jk}, \tag{5.3}$$

where d_{ijk} is the third rank piezoelectric tensor. So when a shock wave impinges on a piezoelectric pressure transducer, the rapid increase in stress induces a corresponding electric displacement, producing a measurable voltage across the crystal electrodes. The transduction process therefore converts a mechanical shock into an electromagnetic signal via lattice-scale charge separation, which can be measured and quantified. [9] [10]

6. LANDING

From the moment the Bell X-1 pierced the sound barrier, aviation entered a regime where velocity, thermodynamics, and wave dynamics become inseparable. An aerodynamic milestone transitioned fluid dynamics into the study of compressibility, and refocused physicists and engineer's focuses on pushing the physical limits of travel. To take the aviation industry into the future and allow for efficient supersonic travel to be brought to the commercial market, we must fully understand the supersonic regime and the dependent properties within it. The purpose of this work has been to build a framework to do so through the use of some example cases.

To develop such framework we first applied Reynolds Transport Theorem to the conserved quantities of mass, linear momentum, and energy in a compressible flowfield. This brought about governing equations, which when applied to one and two dimensional flow allowed

us to study certain phenomena that occur in and around a supersonic vehicle. These being shock waves along the body and heat addition through propulsion devices, such as the ramjet. These phenomena change the flow properties across their respective regions as a function of the upstream Mach number, M_1. Which is the organizing parameter of the flow, linking kinematics, pressure, density, temperature, and entropy in a way that has no analogue in incompressible theory. We then looked at how real world flows can be studied, through the example case of the piezoelectric sensor.

This analysis focused on supersonic inviscid compressible flow of a calorically perfect non-reacting gas, however, only scratches the surface of what is necessary to bring humanity towards the true physical limit of velocity. As the Mach number exceeds $M = 5$, the simplifying assumptions made in this work begin to fail, as viscous effects, chemical reactions, and ionization demand a more complete physical description. Yet the concepts developed here, conservation laws, wave propagation, entropy generation, and thermodynamic relations, form the intellectual foundation upon which hypersonic flow theory is built.

Just as the first supersonic flights revealed the limitations of classical aerodynamic intuition, modern high-speed vehicles continue to challenge our understanding of fluid motion. The study of compressible flow thus remains not only a historical cornerstone of aerospace physics, but an active and evolving frontier, carrying forward the legacy of breaking the sound barrier into regimes still largely unexplored.

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Appendix A: Derivations

1. Entropy

a. Explanation

To understand entropy, we must first understand multiplicity. Imagine that you have 100 coins and that each coin has a perfect 50/50 chance of landing heads or tails. If you were to flip all 100 coins you would get a unique order of heads and tails, this unique order is a *microstate*. You would also have a number of the coins that landed heads up, this number is a *macrostate*. The number of microstates corresponding to a given macrostate is the **multiplicity**. In our example of the coins, there are 101 possible macrostates while there are 2^{100} possible microstates. So the multiplicity of getting 3 heads for example is,

$$\Omega(N, n) = \frac{N!}{n! \cdot (N-n)!} = \frac{100!}{3!97!} = 1.6 \cdot 10^5. \tag{A.1}$$

$1.6 \cdot 10^5$ ways of getting 3 heads. Lets now say that you want to get 50 heads. Then using Equation A.1 we get 1.01×10^{29}, an astronomically bigger number of ways to get 50 heads. So if you were to flip all 100 coins you would expect the amount of coins heads up to be near 50. Now imagine the case of multiplicity for interacting gases, where the number of 'coins' is the number of molecules of gas, this multiplicity would be beyond astronomical near equilibrium and zero, comparatively, near the extrema. This is to say that the more you do an experiment you expect the higher the multiplicity. This notion that multiplicity tends to increase is a form of the *second law of thermodynamics*. Since, however, multiplicities tend to be very large numbers, it is convenient to work their their natural logarithm and define *entropy* to be,

$$S \equiv k \ln \Omega, \tag{A.2}$$

where k is the Boltzman constant which gives entropy, S, units of energy (J/K). We can

know say that the second law of thermodynamics is *entropy tends to increase*. That is that in all processes $ds \geq 0$.

b. Derivation of Equation 2.16

Consider the first law in the form of Equation 2.14. If the heat is reversible and we use the definition of entropy in the form of $\delta q = Tds$, then the first law becomes

$$\begin{aligned} Tds - pdv &= de, \\ Tds &= de + pdv. \end{aligned} \tag{A.3}$$

By taking the definition of enthalpy, $h = E + pv$, and differentiating we find $dH = de + pdv + vdp$. Combining that with Equation A.3 we have,

$$Tds = dh - vdp. \tag{A.4}$$

For a thermally perfect gas, from Equation 2.7, $dh = c_p dT$. Substituting that in we have

$$ds = c_p \frac{dT}{T} - \frac{vdp}{T}. \tag{A.5}$$

Now substituting in the perfect gas Equation (Eq. 2.2) in we have,

$$ds = c_p \frac{dT}{T} - R\frac{dp}{p}. \tag{A.6}$$

Integrating between two states yields,

$$s_2 - s_1 = \int_{T_1}^{T_2} c_p \frac{dT}{T} - R\frac{dp}{p}. \tag{A.7}$$

If we assume that the gas is a calorically perfect gas, where the specific heats are constant, we finally obtain Equation 2.16,

$$s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1}, \tag{A.8}$$

and similarly if we use $de = c_v dT$, the change in entropy can be written as,

$$s_2 - s_1 = c_v \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1}. \tag{A.9}$$

Now, if the process is isentropic than $s_2 - s_1 = 0$. Setting $s_2 = s_1$ we can relate Equations 2.16 and 2.17 directly.

From Equation 2.16,

$$\begin{aligned} 0 &= c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1}, \\ ln\frac{T_2}{T_1} &= \frac{c_p}{R} \ln \frac{p_2}{p_1}, \\ \frac{p_2}{p_1} &= \left(\frac{T_2}{T_1}\right)^{c_p/R}. \end{aligned} \tag{A.10}$$

Recalling Equation 2.12, we can substitute for c_p/R to obtain,

$$\frac{p_2}{p_1} = \left(\frac{T_2}{T_1}\right)^{\gamma/(\gamma-1)}. \tag{A.11}$$

Similarly, from Equation 2.17,

$$\begin{aligned} 0 &= c_v \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1}, \\ \ln \frac{v_2}{v_1} &= -\frac{c_v}{R} \ln \frac{T_2}{T_1}, \\ \frac{v_2}{v_1} &= \left(\frac{T_2}{T_1}\right)^{-c_v/R}. \end{aligned} \tag{A.12}$$

From Equation 2.13, we can substitute for c_v/R to obtain,

$$\frac{v_2}{v_1} = \left(\frac{T_2}{T_1}\right)^{-1/(\gamma-1)}. \tag{A.13}$$

If we recall that $\rho_2/\rho_1 = v_1/v_2$, then we get,

$$\frac{\rho_2}{\rho_1} = \left(\frac{T_2}{T_1}\right)^{1/(\gamma-1)}, \tag{A.14}$$

and putting all equalities together we obtain Equation 2.18,

$$\frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1}\right)^{\gamma} = \left(\frac{T_2}{T_1}\right)^{\gamma/(\gamma-1)}. \tag{A.15}$$

2. Mach Number Introduction

Derivation of Equation 3.24 is as follows. Consider again our one dimensional energy Equation (Eq. 3.15), if we assume that there is no heat addition and that we are using a calorically perfect gas, then Equation 3.15 becomes,

$$c_p T_1 + \frac{u_1^2}{2} = c_p T_2 + \frac{u_2^2}{2}. \tag{A.16}$$

Applying Equation 2.12 this becomes,

$$\frac{\gamma R T_1}{\gamma - 1} + \frac{u_1^2}{2} = \frac{\gamma R T_2}{\gamma - 1} + \frac{u_2^2}{2}. \tag{A.17}$$

Since $a = \sqrt{\gamma R T}$ this becomes,

$$\frac{a_1^2}{\gamma - 1}\frac{u_1^2}{2} = \frac{a_2^2}{\gamma - 1} + \frac{u_2^2}{2}. \tag{A.18}$$

Since we derived this equation for no heat addition, it also holds for an adiabatic flow. Let point 2 be our imagined point where the fluid element is brought adiabatically to Mach 1 at point A. Thus $u_2 = a_2 = a^*$, and we have,

$$\frac{a^2}{\gamma - 1} + \frac{u^2}{2} = \frac{\gamma + 1}{2(\gamma - 1)} a^{*2}. \tag{A.19}$$